

Zili Tan

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RESEARCH INTERESTS

Self-Improving AI Systems, Learning Dynamics, Continual Adaptation, and Neural Architectures.

ACADEMIC BACKGROUND

University of Amsterdam & Vrije Universiteit Amsterdam 09/2025 – 06/2027 (expected)
MSc Physics and Astronomy (joint degree), Theoretical Physics track GPA: 4.0/4.0

Sun Yat-sen University 09/2021 – 07/2025
Bachelor of Science in Physics GPA: 3.7/4.0

PUBLICATIONS & PREPRINTS

Meng Bi, Jinhai Yang, **Zili Tan**, Bowei He, Hong Huang, and Xue Liu. *Wright–Gödel Machine: Mitigating Homogenization in Self-Improving Agents*. **Under review at ICLR 2027**.

RESEARCH EXPERIENCE

Agentic AI Workflows for SMEFT Event Generation at the LHC 10/2026 – Present
MSc Thesis, Advisor: Prof. Wouter Verkerke, Nikhef / University of Amsterdam

- Developing an agentic AI workflow that translates physics-level SMEFT analysis requests into validated Monte Carlo event-generation pipelines using SMEFTsim, MadGraph, Pythia, Delphes, and cluster-based execution.
- Designing tool interfaces, automated failure diagnosis, and physics-aware validation benchmarks based on cross sections, kinematic distributions, reweighting closure, and Wilson-coefficient dependence, with comparisons against expert-produced reference samples.
- Investigating novel training methods for neural simulation-based inference (NSBI) that produce self-calibrated, unbiased estimators by construction, aiming to eliminate the need for post-hoc calibration in statistical measurements.

Genetic Homogenization in Self-Improving Coding Agents 08/2026 – 09/2026
Collaborative project with Meng Bi et al.

- Investigated diversity collapse in recursively self-improving coding agents across Darwin, Huxley, and Mendel Gödel Machine variants.
- Co-introduced genetic homogenization to characterize the loss of heritable design diversity during iterative self-improvement.
- Studied the exploration–exploitation tradeoff underlying this collapse and proposed EXP3 as an alternative to Thompson Sampling for maintaining architectural diversity.

Observer-Relative Inflationary Histories in de Sitter JT Gravity 06/2026 – 09/2026
Advisors: Prof. Ben Freivogel and Prof. Jan Pieter van der Schaar, University of Amsterdam

- Studied de Sitter JT (Jackiw–Teitelboim) gravity coupled to a homogeneous inflaton and a weakly coupled observer.
- Found that the current observer-relative gravitational path integral (GPI) prescription does not normalize the no-boundary state, indicating a possible inconsistency of the Hartle–Hawking proposal in the observer-relative Hilbert space.

Lorentzian Gravitational Path Integral in the de Sitter Static Patch 04/2026 – 06/2026
Advisor: Prof. Jan Pieter van der Schaar, University of Amsterdam

- Surveyed how the Lorentzian path-integral approaches can avoid the conformal factor problem.
- Used Claude Code to expand the Einstein–Hilbert action around the D -dimensional de Sitter static patch.
- Obtained the one-loop spectrum of de Sitter space from the Lorentzian gravitational path integral.

Holographic Cosmology and Inflationary Cosmology 10/2022 – 06/2025
BSc Thesis, Advisor: Prof. Chengcheng Han, Sun Yat-sen University

- Studied the standard cosmological model and inflationary cosmology in Professor Han’s Research Group.
- Reviewed the holographic principle, holographic entropy bounds, and models proposed by Banks and Fischler.

- Analyzed how holographic models address the low-entropy initial condition problem and evaluated the prospects for testing holographic cosmology with future CMB and primordial gravitational-wave observations.

COMPETITIONS & AWARDS

National Training Program of Innovation for Undergraduates

12/2022 – 03/2024

Presenter and Team Member

- Reviewed slow-roll models and compared their predicted power spectra with Planck TT observations.
- The final report was rated **Excellent** (03/2024) in the National Training Program of Innovation for Undergraduates.

China Undergraduate Physics Tournament (CUPT)

12/2021 – 10/2022

Contestant representing SYSU

- Conducted literature reviews on fluid mechanics and Hamiltonian mechanics; designed experiments and built theoretical models for several real-life-inspired physics problems; presented results in the Central South Regional and National rounds.
- Awarded the **National First Prize** (10/2022) of CUPT.
- Awarded the **First Prize** in the Central South Regional Competition (06/2022).

EXTRACURRICULAR ACTIVITIES

SYSU Freshman Mentoring Program

09/2024 – 09/2025

Mentor

- Guided a group of incoming students throughout their first year, helping them adapt to university life and answering their questions about study and campus life.

SYSU Torchwood Association of Physics

10/2022 – 10/2024

Governing Member

- Organized information-sharing seminars and physics contests, including the CUPT Simulation Game.
- Served as a judge in several contests.

SKILLS

- Python, Mathematica and L^AT_EX.